



Personalizing AI art boosts credit, not beauty

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ABSTRACT

While image-generating artificial intelligence (AI) increasingly democratizes art creation, people tend to devalue AI-generated content. Recent work suggests that human use of personalized AI models, trained on a user's past work, can increase credit attributions to the human user for achieving beneficial text-based outputs. We investigated whether this effect extends to visual artistic outputs and further examined the relationship between credit attribution and aesthetic appreciation. Across two studies (N = 774), UK participants evaluated identical paintings that were described as being created either by hand, with a standard text-to-image generative AI system, or with an AI system personalized to the artist. Personalization significantly improved both achievement credit and authorship attribution towards the human user compared to standard AI use. However, it failed to enhance either aesthetic appreciation of the image itself or willingness to categorize the output as "true art"—revealing a striking disconnect between judgments of artistic contribution and artistic value. Our findings suggest that although personalized AI may help bridge the "achievement gap" in credit attribution not only for written works, as demonstrated previously, but also for artistic visual images, it cannot overcome fundamental barriers to aesthetic appreciation of AI art. This challenges assumptions about the relationship between artistic input and aesthetic value, with implications for understanding art categorization and human-AI cooperation in creative pursuits.

1. Introduction

As generative artificial intelligence (GenAI) continues to proliferate through society, a noticeable shift is taking place, from the default use of one-size-fits-all GenAI models trained on massive datasets scraped from the internet, to more *personalized* models that are tailored to individual users' needs, styles, and preferences, or even fine-tuned on their own past outputs (as in Porsdam Mann et al., 2023). Real-world examples of

AI personalization can be found in numerous AI products, such as Google's Gemini, which reportedly absorbs a user's writing style and preferred syntax to write personalized 'smart' email replies on their behalf – a function that also exists in Microsoft Copilot. Taking personalization even further, Delphi AI, a US-based startup, entices users to "clone yourself," stating on their website (<https://docs.delphi.ai/home>) that "Delphi empowers you to create a *digital version of yourself* that thinks and interacts exactly as you do" (emphasis added). "Whether

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you're a creator looking to scale your impact or simply someone who wants to understand yourself better," the ad copy continues, "Delphi enables meaningful, personalized conversations that stay true to your voice."

As the Gemini and Copilot examples illustrate, much of the move toward personalization has focused on written, text-based outputs, or even, as with Delphi AI, making "psychological digital twins"³ of people intended to replicate certain aspects of their personality (Earp et al., 2024a, 2025; Iglesias et al., 2025). However, the personalization trend has also reached artistic creation through the example of AI image generators. Unlike conventional GenAI models that rely solely on text prompts, more advanced models can now take visual inputs, such as selfies, photos of pets, or even scanned or digitized paintings, and recreate them in various artistic styles. Stable Diffusion—a GenAI model developed by Stability AI that boasts 10 million daily users (Katatikarn, 2024)—additionally offers many personalization features, including an 'img2img' function that allows users to input personal images for modification (Andrew, 2023; Andrew, 2024). Even the latest version of the Apple iPhone (version 16 at the time of writing), the best-selling smartphone globally, comes equipped with a "Playground" app that converts photos of friends and family, as directed by user-customized specifications, into artistic images featuring AI-generated avatars that resemble them.

These AI tools are emerging within a landscape of amplified AI use. OpenAI's ChatGPT (which can generate both written and visual content based on text-based prompts) serves over 400 million weekly users (Kant, 2025), while dedicated AI art generators similarly reach millions—Midjourney has 15 million registered users, and DALL-E (OpenAI's image generator) serves 1.5 million daily users (Katatikarn, 2024). A new service, openart.ai, which is targeted specifically at creative artists, makes fine-tuning an AI model on one's own previous artwork, enabling the quick creation of new digital art in one's distinctive style, trivially easy (<https://openart.ai/>). Like other fine-tuning services, it starts with a base model previously trained on vast datasets of relevant material (here, artistic images combined with text-based descriptions of them), and then re-trains the last few layers of the model on a much smaller, user-specific dataset to personalize outputs while preserving the model's general capabilities. "Teach the AI with your best examples," the website instructs, recommending that artists upload "4 to 128 high-quality images to train your model."

As services for personalized AI art creation spread across a large and quickly expanding user base, they raise fundamental questions about who, or what, should be credited as the artist or author of these myriad creations. The user? The company that created the AI? Those whose creations were used (often without their permission) to train the base models prior to user personalization? The models themselves? Questions such as these are being hotly debated (Browne, 2022; Charles, 2025).

In this work, we zoom in on just one of these questions and investigate laypeople's judgments about the answer. Specifically, we ask participants to make judgments about how much credit a *human user* deserves for artistic images produced using a GenAI model, depending on whether or not the model has been *personalized* (i.e., fine-tuned on the user's own past work). Our reason for studying lay attitudes is that any laws or regulations to be developed in this fast-moving area — for example, in relation to copyright assignment, or commercial rights to sell AI-generated images — will need to be informed by the moral sensibilities of the public if they are to be effective, or even seen as legitimate (Smith et al., 2025).

Our reason for focusing on credit attributions to the *human user* of GenAI (as opposed to the AI itself, for instance) is that credit attributions to individual human agents are both psychologically and philosophically the most straightforward to make sense of. In other words, both

laypeople and experts know, with some degree of confidence, what it means to say that a human agent, X, is responsible for bringing about an outcome, Y, such that X deserves at least some amount of credit (or conversely, blame) for bringing about Y (Holborow, 1971). By contrast, it is much less obvious what it means to assign credit to non-human systems such as AI (Lei & Rau, 2021), to corporations, such as AI companies, or to distributed human agents, such as those whose creations have been used to train generative AI systems.

There are, of course, various philosophical accounts of how to make, or make sense of, such credit assignments (e.g. French and Jones, 2022); but they have not commanded universal assent even among philosophers (e.g. Velasquez, 1983), much less members of the general public. Meanwhile, empirical work on laypeople's judgments about, e.g., distributed or corporate responsibility remains nascent (Strohming & Jordan, 2022). Since we aim to study laypeople's credit judgments in relation to an *even more nascent* phenomenon, namely, the production of artistic images using personalized versus standard GenAI tools, we have decided to focus on credit attributions to the human user — rather than various other parts of the human-AI-corporation complex — to enable greater conceptual clarity (as well as interpretability of empirical findings) in this initial line of work.

Such a focus is also, we note, consistent with philosophical accounts according to which AI, whether personalized or otherwise, does not currently meet criteria to qualify as an artistic collaborator due to its (widely assumed) lack of mental states and inability to refuse participation among other factors (Anscomb, 2025). Thus, the creative relationship between humans and AI should not, on such accounts, be described as an artistic collaboration, but rather as *AI-assisted production* of artistic objects based on human prompting (Anscomb, 2025). Such accounts, of course, may be contested (Bayindir & Danaher, 2025).

Credit attributions to human users of *personalized* AI have not, to our knowledge, yet been empirically studied in relation to the visual arts domain (for a study on perceptions of *written* outputs created with standard versus personalized AI, see Earp, Porsdam Mann, et al., 2024). However, multiple studies have documented public attitudes toward AI-assisted achievements (including artistic achievements) using *standard* models, yielding a range of potentially relevant findings. In a typical pattern of results, when human users are described (in an experimental setting) as using an AI tool to produce a positive outcome, they receive diminished attributions of credit or authorship compared to unassisted creation. Using a standard large language model (LLM) to generate beneficial written content, for example, reduces credit attributed to the human user, compared to if they write the material themselves — an effect that has been replicated across four different countries (Earp, Porsdam Mann, et al., 2024).

Moreover, when a person is seen as 'outsourcing' certain tasks to AI, particularly creative tasks with perceived socio-relational value, participants think more "negatively about not only the person and their motivations, but also the outsourced work itself" (Claessens et al., 2025). For example, when visual artwork is described as being AI generated, not only is less effort attributed to the human user (Bellaiche et al., 2023), but the artwork itself is seen as less creative and original (Hong & Curran, 2019; Magni et al., 2023). Similar effects have been observed for music and poetry (Epstein et al., 2020; Khosrowi et al., 2023).

On the other hand, when a creative work is *collaboratively* produced — such as an AI-generated but human-curated haiku in one study — the work itself may be seen as more beautiful (Hitsuwari et al., 2023), even though people cannot otherwise reliably distinguish human- versus AI-generated poetry (Köbis & Mossink, 2021). Although such findings are preliminary, and we discuss some issues with their interpretation below, they could suggest that greater human involvement in a human-AI jointly produced artistic output might not only increase credit attributions to the human user but also boost aesthetic appreciation of the output itself.

Accordingly, in this work, we assess whether *personalization* of

³ Also called AI Simulations of Individual Minds or AI SIMs (Voinea et al., 2025)

GenAI, specifically by means of fine-tuning a model on the previous creative outputs of a single human user, does indeed increase attributions of credit to the user for newly generated model outputs. We also assess the impact of such personalization, if any, on aesthetic judgments of the outputs themselves.

1.1. Personalization and credit attribution

Why might personalization be expected to influence credit attributions to human users for AI-assisted content production? To formulate a hypothesis about this, we must first understand why standard, non-personalized AI use seems to undermine credit attributions to human users. On standard philosophical accounts, agents should receive credit for achievements that demonstrate their ability to bring about certain outcomes, not by any means whatsoever, but rather, by certain *appropriate* (e.g., intrinsically or instrumentally valuable, or praiseworthy) means (Bradford, 2015). For example, a climber who reaches the summit by exerting themselves, one step at a time, for several hours, deserves more credit for this achievement than a ‘climber’ who reaches the summit by helicopter in just a few minutes (Gordon et al., 2025). While the first individual “earns” their achievement, in part by employing such admirable qualities as grit and perseverance, the second individual, despite reaching the same destination, has taken a shortcut – one that undermines the creditworthiness of getting there.

Building on this insight, Danaher and Nyholm (Danaher & Nyholm, 2021) have identified four specific factors that, they argue, should undergird the evaluation of human achievements: the *cost* to the agent of their commitment to producing the outcome; the nature of the *causal contribution* of the agent in bringing the outcome about; the *voluntariness* of the agent’s efforts in doing so; and the *value* of the outcome itself. Thus, under typical conditions, the more time, effort, attention, or stress exerted by an agent; the stronger and more direct the agent’s causal contribution; the more voluntary and deliberate their efforts; and the more valuable the outcome produced, the more credit the agent should get for the achievement.

From this account (Danaher & Nyholm, 2021), we can see why standard AI use may undermine the value of an achievement in several ways. First, AI systems significantly reduce the time, effort, attention, and stress required from human users to bring about certain outcomes (that is, the cost of their commitment). Second, the causal contribution of the user may be diminished on two fronts: not only can users bring about a seemingly valuable outcome with minimal effort (e.g., typing a short prompt into the computer to generate a beautiful visual art piece), thus reducing the strength of their causal contribution, but the causal relationship between their input and the outcome in question is mediated by innumerable factors (e.g., the datasets used to train the model, the model’s programming, its ‘temperature’ setting, and so on), making their causal contribution less direct and more diffuse. And since the model may produce an outcome that is rather different from what the user expected or intended, the voluntariness of bringing the outcome about (though not, presumably, the voluntariness of using the model in the first place) is also called into question.

Finally, in the domain of art, especially, even the perceived value of the output may be reduced, since part of what people value in art is precisely the means by which it was created – that is, its causal history and connection to the artist (compare the perceived value of an original painting to that of a physically indistinguishable forgery, for instance) (Newman & Bloom, 2012). For all these reasons, it should be unsurprising, then, that human users tend to receive less credit for outcomes produced with GenAI compared to similar outcomes produced without AI assistance.

What about *personalizing* AI, then? How should that influence credit attribution? Drawing on the same four criteria from Danaher and Nyholm (Danaher & Nyholm, 2021), we can see that credit attributions should increase for personalized AI use in several ways, compared to standard AI use, when generating an artistic image. (By ‘should’

increase, we mean from a philosophical or normative point of view; whether they do increase, in terms of laypeople’s judgments, is what we are investigating in this work).

First, by being fine-tuned on a human user’s past artistic work, the AI model inherits, or is influenced by, the time, attention, effort, and stress which *that specific user* put into the creation of the original artwork. Outputs produced by the model will therefore reflect the user’s (previously demonstrated) costly commitment. Indeed, by virtue of the way the model’s internal weights must change if the fine-tuning process is successful, the causal relationship between the user’s input and the model’s output is also strengthened. The voluntariness of the outcome may also increase. For example, if the user previously painted cubist works in a ‘signature’ style, using specific colours, composition, and symmetry, and intends for their new AI-generated artworks to reflect those distinctive features, this will be much more likely if they use a personalized AI than if they use a standard one. Finally, insofar as those same valued features do end up characterizing the new personalized AI-generated images and would not have done so had the images been created using standard AI, then the images *themselves* may be more valuable than they would have been otherwise, resulting in a greater overall achievement for the user.

Consistent with this theorizing, but in the text-based rather than visual domain, a recent empirical study by Earp et al. (2024) found that hypothetical users of a personalized large language model (LLM) received more credit for valuable written content, compared to the same content described as being produced through the use of a standard or “off-the-shelf” LLM such as OpenAI’s ChatGPT. Many participants reasoned that the personalized model reflected the user’s *previous* effort, skill, or creativity, and thus substantially preserved their creditworthy input in bringing the new written content about.

1.2. Personalization and aesthetic value

How might personalization affect *aesthetic* judgments, then? As mentioned earlier, one possibility is that it could also increase participants’ aesthetic appreciation of model outputs compared to the same outputs made with standard, non-personalized AI. Alternatively, the well-documented “algorithm aversion” effect — the tendency to devalue content known to be AI-generated (Epstein et al., 2020; Hong & Curran, 2019; Khosrowi et al., 2023) — demonstrates that involvement of an AI in generating an artistic output dampens aesthetic appreciation of the output, compared to physically identical output believed to be human-made (Bellaiche et al., 2023). This effect might be robust enough that it holds *regardless* of whether the AI in question happens to be personalized or standard. We consider both possibilities in this section.

First, however, we must ask how *identical* outputs could result in different aesthetic judgments in the first place. If two pieces of art look exactly the same, how could one — even in principle — be judged as more beautiful than the other? To answer this question, a distinction must be drawn between two schools of thought in aesthetics: *aesthetic empiricism* and *aesthetic contextualism*.

Aesthetic empiricism holds that the aesthetic properties of an artwork depend solely on its perceptual properties. In contrast, aesthetic contextualism holds that the aesthetic qualities of a work depend not only on perceptual properties but also on the historical context of the artwork: where it comes from, who created it, and how it came about. Under this view, even two perceptually indistinguishable artworks can differ in their aesthetic value, as this is influenced by the artwork’s “origin story” — which may include certain facts about the process by which the artwork was created, such as the nature or extent of the artist’s contribution (Chamberlain et al., 2018). Contextualists thus maintain that art experience, categorization, evaluation, and criticism are inseparable from the historical circumstances of creation, including the creator’s intentions, inputs, and expectations (Danto, 1981; Levinson, 2017; Walton, 1970).

Even ordinary people who are not versed in art theory seem to

evaluate artworks in this ‘contextualized’ way: not as isolated objects but as ‘endpoints of a performance’ (Dutton, 2003) — meaning the creative process itself, and the identity of the creator, contribute to aesthetic value (recall the ‘original versus forgery’ example from before). When AI systems handle substantial portions of the creative process, observers may experience diminished aesthetic responses because the work’s origins are less directly tied to any specific expression of human agency, intention, or creativity. These factors, in turn, might be thought to constitute much of the ‘point’ of art — that is, what gives art its distinctive value, or even what makes something *art* to begin with. This would suggest a parallel to the “achievement gap” in credit for AI-generated outputs: just as users receive less *credit* for AI-assisted work, observers may find such work less *aesthetically* compelling, perhaps due to the perceived lack of human agency or involvement.

How might *personalization* make a difference to this prediction? We return to the question with which we started. On one view, personalizing AI art should both increase credit attributions to the human user *and* boost aesthetic appreciation of the resulting artwork. In the personalized case, since model outputs would be more closely tied to the expressive intentions, creative style, effort, and ability of a specific human user, perhaps the ‘AI penalty’ on aesthetic judgments of the output should likewise significantly decrease. In other words, perhaps personalizing AI would address some of the main sources of people’s relative dispreference toward AI-generated art, such as its lower perceived creativity (Bellaiche et al., 2023), expressiveness (Magni et al., 2023), or personal style (Hong & Curran, 2019). If so, we should then expect that personalization would improve aesthetic judgments of outputs compared to those created with standard GenAI.

Empirical support for this possibility comes from the ‘haiku’ study described earlier, in which human-curated AI poems were judged as more beautiful than randomly selected AI-generated poems (Hitsuwari et al., 2023). In that study, the authors sought to determine whether a greater level of human “involvement” in AI-generated poetry would increase — not the credit attributed to the human user — but rather, the *aesthetic responses* of study participants to the AI-generated poetry itself. And at face value, that is precisely what they found. If we accept the authors’ interpretation of their results, and we consider AI personalization as likewise increasing the “involvement” of a human user in the output (at least, compared to using standard AI), then perhaps the artistic outputs of a *personalized* AI will similarly elicit higher beauty ratings from participants.

However, there is a problem with this reasoning. In the haiku study, the authors did not take *identical* haikus and manipulate participants’ perceptions of whether there was more or less human involvement in the process of creating those haikus; rather, the authors selected a subset of AI-generated haikus that had already been “highly rated for their beauty by three amateur/novice haiku artists” (Hitsuwari et al., 2023) and classified those specific haikus as “human in the loop” or HITL haikus, representing a high level of human intervention (i.e., human *selection* or *curation* of haikus that were generated entirely by AI). Meanwhile, they classified a *different* subset of randomly selected AI-generated haikus as “human out of the loop” or HOTL haikus, representing a lower level of human intervention. Given that the HITL haikus were selected for inclusion in the study precisely because they were previously rated as being especially beautiful (i.e., during pretesting), it is not surprising that they would then also be rated as more beautiful during the study itself.

Thus, at this point, it is *empirically* uncertain whether greater human “involvement” in a process of AI art creation — whether by means of personalization or by other means — results in greater aesthetic appreciation of the output itself, *holding the output constant*. In the present work, we try to account for this issue by presenting identical artworks (in this case, paintings rather than poems) between conditions, while manipulating *only* the description of how the artworks were produced: i.e., by hand (no AI use), by using a standard AI trained on images from the internet, or by using a personalized AI, fine-tuned on a

series of scanned and digitized paintings the user had previously made by hand.

In addition to empirical issues, however, there are also theoretical reasons to question whether AI personalization will influence aesthetic judgments. Given the strong evidence on aversion to machine authorship cited previously, personalization may have little effect on aesthetic judgments, insofar as it does not change the fundamental identity of the creator — AI remains the immediate generator of the artwork. From this perspective, regardless of the type of GenAI (personalized or standard), AI-generated works would receive lower aesthetic judgments due to the primary role of the generative AI in carrying out the creation process.

Finally, a similar question arises as to whether this ‘algorithmic aversion’ might influence not only people’s aesthetic ratings of AI-generated images, but also their willingness to classify such images as art in the first place. In other words, might people call into question the very status of the images as art? Researchers have begun to explore how people categorize AI-generated images (Mikalonytė & Kneer, 2022). A recent study found that though people are willing to categorize paintings and musical works produced by AI-driven robots as art, they are less willing to describe the robots who produce them as artists (Wu et al., 2020). To our knowledge, however, there is no prior work investigating the effect of AI personalization on the basic categorization of images as art.

To summarize, we have the following expectations. First, drawing from the achievement gap framework (Danaher & Nyholm, 2021) and recent empirical findings (Earp, Porsdam Mann, et al., 2024), we hypothesize that personalizing AI art should improve credit attribution to the human user compared to standard AI use, since personalization strengthens the relationship between the output and creditworthy inputs from the user. Second — although the effects of personalization on aesthetic judgment remain theoretically ambiguous — given the consistent findings that any AI involvement reduces aesthetic appreciation, we hypothesize that both personalized *and* standard AI use will receive lower aesthetic ratings than unassisted human creation.

We test these predictions across two studies. Study 1 takes an exploratory approach to examine personalization effects across multiple outcome measures (credit attribution, commercial rights, aesthetic appreciation), allowing us to assess the breadth and robustness of theoretically predicted effects in visual art creation. Study 2 then provides a pre-registered confirmatory test of our key predictions while examining the underlying mechanisms (effort, ability, creativity attributions) that theory suggests should drive any personalization effects.

2. Study 1

In Study 1, we explored whether attributions of credit, commercial rights, and aesthetic value can be influenced by AI personalization. Specifically, in addition to measuring credit attributions, we also assess judgments about the creator’s ‘right to sell’ the artwork, as commercial rights represent a concrete application of credit attribution — if someone deserves credit for creating an artwork, it follows that they would have rights to profit from it. Second, we measure aesthetic appreciation to test whether personalization can overcome the documented bias against AI-generated art. If personalized AI preserves more human agency in the creative process, it may also preserve more of the aesthetic value that observers typically associate with human creativity.

2.1. Method

2.1.1. Open science

The materials, anonymised data, and analysis code for this study are available on the Open Science Framework (OSF) (<https://osf.io/hza9g/>). As this was an exploratory study, it was not pre-registered.

2.1.2. Design

In a between-subjects design, participants were randomly assigned to

one of 3 (text-to-image model [standard, personalized] and control) conditions. In each condition, participants were randomly presented with one of 5 paintings, and paintings were treated as a random effect. The paintings and the prompts used to generate them can be found in Fig. 1 (the AI watermark was removed before showing paintings to participants).

In the AI conditions, Robin, a fictitious character, was described as using either (a) a personalized text-to-image AI model (fine-tuned on 25 paintings Robin had previously made by hand), or (b) a standard text-to-image AI model (trained on a huge corpus of images and corresponding text descriptions from the internet) to create a painting. In the control condition, Robin was described as creating the new painting by hand. In all conditions, Robin was described as having previously created 25 hand-made paintings, to ensure consistency in the character's baseline competencies, and, at the end of the vignette, as having produced a novel image similar to what they had in mind – to control for intention or 'voluntariness' of the outcome – with one of the 5 paintings accompanying the vignette. See Supplementary Material Section 1 for the precise wording of the vignettes.

The 5 paintings used in this study were selected from an original set of 10 paintings generated by the research team using Open AI's image generator, Dall-E 2, in the first week of November 2023. This selection was based on a pre-test that evaluated two criteria: AI believability (assessed by asking "How believable is it that this painting was made by an AI?"), overall quality – which was a combination of aesthetic value judgments (assessed by asking "How much do you like this painting?") and behaviour intention judgments (assessed by asking "Would you buy this painting?"). The paintings scoring highest on both measures were chosen for the study. Pre-test results are in Section 1 of the Supplementary Material.

After reading the vignette, participants responded to our main dependent variables, rating their attributions of credit towards Robin and their aesthetic appreciation of the painting they saw. They also rated

their emotions when viewing the artwork, which was always visible at the top of the page throughout the survey, after its first appearance.

2.1.3. Participants

We collected UK survey data from 251 participants through Prolific in April 2024. We applied a comprehension check assessing participants' understanding of how text-to-image AI models work. After reading the explanatory vignette, participants were asked to select the correct answer describing how a text-to-image output is produced. We excluded data from 21 participants who failed this check.

This gave us a final sample of 230, including 111 women, 114 men, 3 non-binary participants, and 2 participants who preferred not to state their gender ($M_{age} = 46.45$, $SD_{age} = 12.63$).

Our sample reported generally low levels of experience with LLMs ($M = 32.12$, $SD = 28.89$ on a 100-point scale) and with text-to-image GenAI ($M = 14.4$, $SD = 23.24$ on a 100-point scale). They also reported low art interest ($M = 2.70$, $SD = 1.26$ on a 7-point scale); exact items described below.

2.2. Measures

2.2.1. Credit

Participants were asked to rate how much credit Robin should receive for the painting on a scale of 0 (none at all) – 100 (all of it).

2.2.2. Right to sell

Participants were also asked whether they thought Robin should be allowed to sell copies of the painting for personal profit, rated on a scale of 0 (definitely not) – 100 (definitely yes). They were also given a chance to provide a short open response explaining their answer to this and the previous question (these data were not analysed for the present study). Credit and right-to-sell items were presented in a randomised order.



Fig. 1. Final Stimuli. AI-generated images used in studies 1 and 2. *Note.* These five paintings were generated by our team using Dall-E by entering the following prompts: (A) "A John Hopper style painting of a street in Karachi where children are playing"; (B) "People standing around bonfire by the lake, someone dancing, oil painting"; (C) "Oil painting, bleak colours, astronomical observatory on the hill"; (D) "Please help me generate a picture, the requirements are: Draw a still life painting that resembles one created by a human artist, featuring relatively rough brushstrokes"; (E) "Bridge joining the earth and the moon, painting, wide brushstroke". The AI watermark was removed from the paintings before showing them to participants. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

2.2.3. Aesthetic appreciation

Participants were asked to rate their agreement on a scale of 1 (strongly disagree) – 7 (strongly agree) with three statements presented in random order as follows: “I liked this painting,” “I found this painting beautiful,” and “I found this painting aesthetically pleasing.” These items – beauty, liking, and pleasure, often referred to as ‘hedonic tone’ (Berlyne, 1974) – are regularly used in empirical aesthetics. The statements were adapted from the scale used by Marin and colleagues (Marin et al., 2016).

We calculated an overall aesthetic appreciation score by taking the mean of participants’ responses to these items. Reliability analyses indicated that the items were almost perfectly correlated ($\alpha = 0.98$) implying some degree of empirical redundancy between the items. Previous research has shown that these three measures are often highly correlated (Marin et al., 2016; Skov & Nadal, 2020). However, as the concepts of liking, beauty perception, and aesthetic pleasure are theoretically distinct (Briellmann & Pelli, 2019), we chose to retain all three items in the combined measure.

2.2.4. Aesthetic emotion

Participants were also asked to rate how much they experienced each of a set of 11 emotions when viewing the painting on a scale of 1–7 (not at all to very). The aesthetic emotions scale was adapted from Schindler et al. (Schindler et al., 2017). Results for this exploratory measure are included in the [Supplementary Material Section 3](#).

2.2.5. Demographics and individual difference measures

Measures of participant age, gender, experience with AI, art interest (adapted from the Vienna Art Interest and Art Knowledge Questionnaire; Specker et al., 2020) and art expertise were also collected. In Section 2 of the Supplementary Material we report the primary analyses with the inclusion of these measures as covariates.

2.3. Results

A linear mixed model was fitted using restricted maximum likelihood (REML) estimation to examine the effect of condition on credit, right to sell, and aesthetic appreciation scores. The model included a fixed effect for condition and a random intercept for painting. *P* values following multiple comparisons are reported with Tukey correction.

2.3.1. Examining credit attribution and right to sell judgments across conditions

Personalizing AI significantly elevated credit scores ($M = 45.6$, 95 % CI [40.0, 51.2]) compared to standard AI ($M = 31.8$, 95 % CI [26.2, 37.4], $\Delta M = 13.8$, 95 % CI [4.6, 23.0], $t(223) = 3.53$, $p = .002$, Cohen’s $d = 0.51$, 95 % CI [0.19, 0.83]). Additionally, when Robin made the painting by hand ($M = 87.3$, 95 % CI [81.7, 92.9], $t(227) = 31.86$), higher credit scores were given than when using either personalized AI ($\Delta M = 41.7$, 95 % CI [32.5, 50.9], $t(227) = 10.70$, $p < .001$, Cohen’s $d = 1.81$, 95 % CI [1.4, 2.2]), or standard AI ($\Delta M = 55.5$, 95 % CI [46.3, 64.7], $t(224) = 14.20$, $p < .001$, Cohen’s $d = 2.64$, 95 % CI [2.2, 3.1]).

Each of these pairwise differences can also be expressed as a *common language effect-size* (CLES) (Zhou & Kawabata, 2023), i.e., the probability that values in one condition (e.g., personalized AI) will be higher than the values for the same measure in another condition (e.g., standard AI) when repeatedly sampling pairs of values. Applying this method, credit attribution for personalized AI use was higher than credit attribution for standard AI use in 64 % of comparisons. On the other hand, credit attribution for hand-made paintings was higher than attributions for personalized AI use in 89 % of cases, and for standard AI use in 95 % of cases. This demonstrates that while AI use in generating an artistic

image reduces credit attribution to the human artist compared to making the painting by hand, personalized AI use can preserve credit attribution relative to standard AI use.

Similar to the pattern for credit, personalized AI use significantly elevated Robin’s right-to-sell scores ($M = 63.2$, 95 % CI [56.2, 70.2]) compared to the use of standard AI ($M = 50.1$, 95 % CI [43.1, 57.1], $\Delta M = 13.1$, 95 % CI [1.6, 24.6], $t(227) = 2.69$, $p = .020$). When Robin made the painting by hand ($M = 88.8$, 95 % CI [81.8, 95.7]), higher right-to-sell scores were given than when using either personalized AI ($\Delta M = 25.6$, 95 % CI [14.2, 36.8], $t(227) = 5.29$, $p < .001$) or standard AI ($\Delta M = 38.7$, 95 % CI [27.2, 49.9], $t(224) = 7.98$, $p < .001$). Expressed as a CLES, right-to-sell scores for hand-made paintings were higher than those for standard AI use in 82 % of comparisons, and for personalized AI use in 76 % of comparisons. Finally, the scores for personalized AI use were higher than those for standard AI use in 60 % of comparisons.

These results follow the same pattern as credit, whereby Robin’s perceived right to sell dropped in both AI conditions, but personalized AI use elevated these scores relative to standard AI use (see [Fig. 2](#)).

2.3.2. Aesthetic appreciation and AI authorship

Unlike our prior findings, personalization did not significantly influence aesthetic appreciation. Robin’s paintings received similarly low aesthetic appreciation scores in the personalized AI condition ($M = 55.5$, 95 % CI [47.5, 63.5]) and the standard AI condition ($M = 52.9$, 95 % CI [44.8, 60.9]), with the difference in scores between the two AI conditions being very small and statistically non-significant ($\Delta M = 2.63$, 95 % CI [−6.6, 11.9], $t(226) = 0.68$, $p = .78$). By contrast, Robin’s hand-made paintings ($M = 64.1$, 95 % CI [56.0, 72.1]) received significantly higher aesthetic appreciation scores than paintings made with standard AI ($\Delta M = 11.19$, 95 % CI [1.9, 20.5], $t(228) = 2.85$, $p = .013$, Cohen’s $d = 0.46$, 95 % CI [0.1, 0.8]). Paintings made with personalized AI received lower aesthetic appreciation scores than Robin’s hand-made paintings; however, this difference was statistically non-significant ($\Delta M = 8.56$, 95 % CI [−0.7, 17.8], $t(227) = 2.18$, $p = .076$, Cohen’s $d = 0.41$, 95 % CI [0.1, 0.7]).

Expressed as a CLES, aesthetic appreciation scores for hand-made paintings were higher than those for paintings made with standard AI in 63 % of comparisons, and higher than those for paintings made with personalized AI in 63 % of comparisons. Meanwhile, aesthetic appreciation scores for paintings made using personalized AI were higher than those for paintings made using standard AI on 52 % of comparisons (in other words, near chance, reflecting a non-significant difference). These results suggest that personalization is ineffective in fostering aesthetic appreciation of AI-generated art (see [Fig. 3](#)).

2.4. Discussion

Our exploratory study reveals two key findings. First, personalized AI use elevates both credit attribution to a human artist and their perceived right to sell an artistic image compared to standard AI use. This is consistent with previous findings (Earp, Porsdam Mann, et al., 2024) on the potential for personalization to improve credit attribution in relation to written content and extends these findings, for the first time, to the visual art domain.

On the other hand, we find that the use of personalized AI, compared to standard AI, to generate an artistic image fails to significantly elevate aesthetic appreciation, suggesting that there might be an AI-aversion effect in relation to aesthetic judgments that personalization may not mitigate. Furthermore, since the same 5 paintings appear in all AI and control conditions, our findings are consistent with previous evidence that people like AI-made art less when it is believed to have been made with the help of an AI (Bellaiche et al., 2023; Chamberlain et al., 2018;

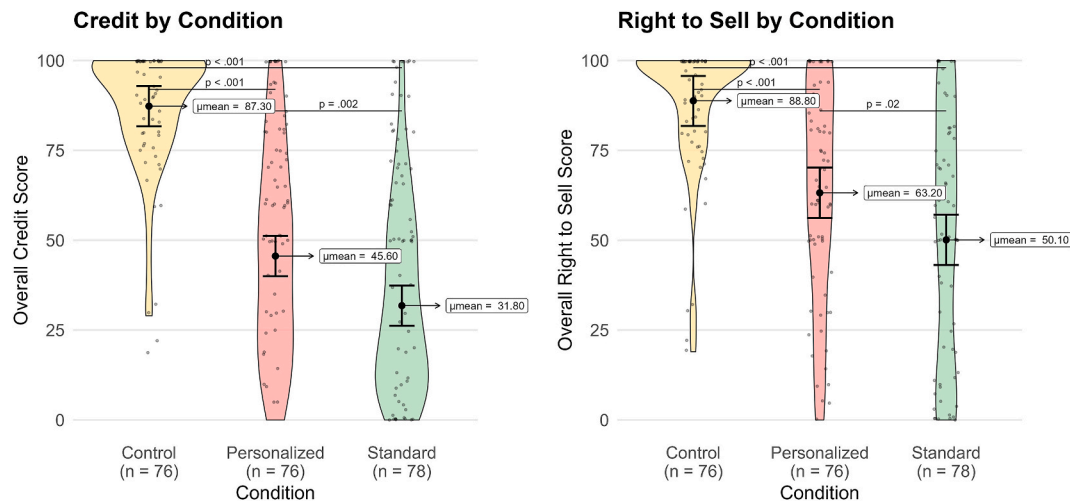


Fig. 2. Attribution of Credit and Right to Sell across Experimental Conditions. Violin plots showing participants' ratings for credit attribution (left) and right to sell (right) across conditions in Study 1.

Hong & Curran, 2019; Magni et al., 2023; Newman et al., 2014; Wu et al., 2020), and not because of any visual property of the artwork itself.⁴

3. Study 2

Given the novelty of the findings and the exploratory nature of Study 1, we decided to conduct a pre-registered replication study, with sample size estimation based on an *a priori* power analysis. To extend our preliminary findings, we further explored the criteria that may be used to evaluate (i) how much credit a user deserves for an output, and (ii) the aesthetic value of the paintings. Drawing from prior work (Bellaiche et al., 2023; Magni et al., 2023; Newman et al., 2014) on effort heuristics and research on perceptions of AI-generated outputs, we also measured participants' attributions of effort, ability, and creativity. Finally, in order to examine the extent to which personalization might (fail to) improve aesthetic value, we directly assessed, with an exploratory measure, whether people even consider AI-generated images to be truly art.

Our pre-registered hypotheses were as follows:

H1. Use of a personalized generative AI to create an artistic image elevates credit attribution to the human user—relative to use of standard AI.

H2. Use of any kind of AI, compared to unassisted human creation, to generate an artistic image reduces aesthetic appreciation of the image.

⁴ One possibility, as discussed in the Introduction to this paper, is that they think the way an artistic image comes about (as opposed to solely the visually perceptible, physical properties of the image) make a difference to its aesthetic value, or even to its status as being truly art. To extend this point, we note that in handmade art, unlike AI-generated art, the resulting artworks have physical contact with the creator, which could also be a factor in our findings. If so, our results would be consistent with findings in the literature on “contagion,” which show that people often believe an object can be, as it were, contaminated with the person's essence through physical touch, leading, in negative cases, to an aversion to touching objects that have been in contact with morally blameworthy individuals (Bates et al., 2015; Newman et al., 2011). But there are positive cases as well. For example, this effect extends to the way people positively evaluate artworks. Evaluations tend to be higher when there has been physical contact between the artwork and its creator, as physical contact is thought to allow the artwork to contain part of the creator's essence (Newman & Bloom, 2012; Schuirmann, 1987).

3.1. Methods

3.1.1. Open science and power analysis

Data, pre-registrations of sample sizes and primary analyses, and Supplementary Materials are available on the Open Science Framework here (<https://osf.io/hza9g/>).

Using the *TOSTER* package in R, we estimated the sample size to detect equivalence between the personalized and standard conditions (within the bounds of Cohen's $d = 0.30$) using the “two one-sided tests” (TOST) procedure (Schuirmann, 1987) and setting the alpha level to .05 and power to 90 %. This resulted in a target sample size of 723 (241 per condition).

We also estimated the sample size to detect a difference (of Cohen's $d = 0.46$; based on Study 1 data) between the standard and control conditions using the *pwr.t.test* function in the *pwr* package, which resulted in a target sample size of 303. Taking the higher of the two sample size estimations (target $N = 723$), we preregistered that we would collect data from 723 UK participants.

3.1.2. Sample

As pre-registered, we collected data from 723 UK participants through Prolific in July 2024. We excluded data from participants who did not pass comprehension checks.

The first comprehension check assessed participants' understanding of how text-to-image AI models work. After reading the explanatory vignette, participants were asked to select the correct answer describing how a text-to-image output is produced. We excluded data from 48 participants who failed this check.

The second check, administered after participants read the experimental scenario, assessed their understanding of the specific condition to which they were assigned. This manipulation check asked participants to identify how Robin—the artist in the vignettes—produced the painting. Response options included: making the painting by hand, producing the painting using a standard AI, and producing the painting using a fine-tuned AI trained on Robin's own past paintings (see the Supplementary Materials for full text). We excluded data from 131 participants who failed this check. Finally, two participants' data were further excluded because they did not agree to pay attention.

This gave us a final sample of 544 participants, including 277 women, 263 men, and 4 non-binary participants ($M_{age} = 46.75$, $SD_{age} = 15.83$). Our sample reported low levels of experience with LLMs ($M = 34.35$, $SD = 30.20$ on a 100-point scale) and with text-to-image GenAI ($M = 14.72$, $SD = 23.38$ on a 100-point scale). They also reported low art interest ($M = 2.87$, $SD = 1.27$ on a 7-point scale), and only 19 participants said ‘yes’ to having art expertise.

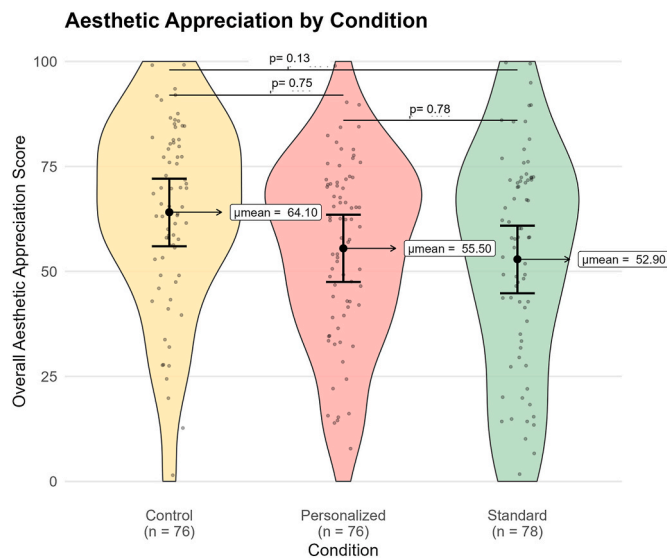


Fig. 3. Aesthetic Appreciation Scores. Violin plots showing participants' ratings for aesthetic appreciation across conditions in Study 1.

3.2. Measures

3.2.1. Credit

While Study 1 examined credit and right to sell as separate measures, Study 2 used a more comprehensive credit attribution score combining three items (rated 0–100). These items showed high reliability ($\alpha = .92$) and were averaged into a single credit attribution score:

1. How much credit does Robin deserve for this painting? Rated from 0 (none at all) to 100 (all of it)
2. Should Robin be allowed to sell copies of this artwork for personal profit? Rated from 0 (definitely not) to 100 (definitely yes)
3. Should Robin be named as the author of this artwork? Rated from 0 (definitely not) to 100 (definitely yes).

3.2.2. Aesthetic appreciation

We used the same three aesthetic appreciation items as in Study 1 (liking, beauty, and aesthetic pleasure) but measured them on a 0–100 scale rather than the 7-point scale used previously. As in Study 1, these items showed very high reliability ($\alpha = .97$) and were averaged into a single score.

Further to these measures, which were included in Study 1, in Study 2 we added several new measures to examine potential mechanisms behind these judgments:

3.2.3. Effort

We asked participants to rate Robin's effort on a scale of 0–100, across three items. Item 3 was reverse scored. We averaged these three items to compute a final effort score ($\alpha = .76$).

1. How much effort did Robin put into making this painting? Rated from 0 (no effort at all) to 100 (a lot of effort)
2. How easy or difficult was it for Robin to produce this painting? Rated from 0 (extremely easy) to 100 (extremely difficult)
3. How trivial was it for Robin to produce this painting? Rated from 0 (not trivial at all) to 100 (extremely trivial).

3.2.4. Ability

We asked participants to rate Robin's ability on a scale of 0–100 across three items. We averaged these to compute a final ability score ($\alpha = .95$).

1. How much skill did Robin show in making this painting? Rated from 0 (no skill at all) to 100 (a lot of skill)
2. How much did this painting require talent on Robin's part as an artist? Rated from 0 (no talent at all) to 100 (a lot of talent)
3. Did this painting require expertise on Robin's part to produce? Rated from 0 (no expertise) to 100 (a lot of expertise)

3.2.5. Creativity

We also asked participants to rate Robin's creativity across three items on a scale of 0–100. We averaged these to compute a final creativity score ($\alpha = .84$).

1. How much creativity did Robin express in making this painting? Rated from 0 (no creativity) to 100 (complete creativity)
2. Would you say this painting reflects Robin's own unique style as an artist? Rated from 0 (definitely not) to 100 (definitely yes)
3. How original to Robin is this painting? Rated from 0 (completely unoriginal) to 100 (completely original).

3.2.6. Art categorization judgments

Finally, we asked participants whether the painting could be considered true art across four items rated on a scale of 0–100. Item 3 was reverse scored. We averaged these to compute a final "art categorization" score ($\alpha = .91$).

1. Do you consider this image a true work of art? Rated from 0 (not at all) to 100 (definitely)
2. Would you classify this image as belonging to the category *art*? Rated from 0 (not at all) to 100 (definitely)
3. However pleasing this image may be to look at, it isn't really art. Rated from 0 (completely disagree) to 100 (completely agree)
4. This is an artistic image. Rated from 0 (completely disagree) to 100 (completely agree).

3.2.7. Demographics, individual difference, and exploratory measures

Demographics and individual difference measures were the same as in Study 1. At the end of the survey, as an exploratory measure, participants were additionally asked to rate their agreement with statements attributing the following properties to AI: Sentience, autonomy, subjective experience, consciousness, and well-being on a scale of 0 (completely disagree) to 100 (completely agree). See Supplementary Materials Section 2 for the exact items, and Section 4 for the associated exploratory analyses.

3.3. Design

We used the same design and materials as Study 1, with participants randomly assigned to one of three conditions (standard AI, personalized AI, or control) and shown one of the five pretested paintings.

After reading the vignette, participants responded to our main dependent variables, rating their attributions of credit towards Robin and their aesthetic appreciation of the painting they saw. They also responded to our exploratory variables, rating their perceptions of Robin's effort, ability, and creativity, and whether they perceived the painting as true art (full items described above). All items were rated on a scale from 0 to 100. The painting was always visible at the top of the page as participants responded to the questions.

3.4. Results

In accordance with our pre-registration, we conducted two linear mixed effects regressions, with each of our dependent variables. For both regressions, condition was set as the independent variable, and painting was set as a random effect. All analyses were performed using the lme4 package in R (Bates et al., 2015; R Core Team, 2023). *P* values following multiple comparisons are reported with Tukey correction.

3.4.1. Is credit attribution elevated for personalized AI use?

Consistent with H1 and the results of Study 1, use of a personalized AI ($M = 54.6$, 95 % CI [50.0, 59.2]) versus standard AI ($M = 38.1$, 95 % CI [33.0, 43.1], $\Delta M = 16.6$, 95 % CI [10.1, 23.0], $t(540) = 6.06$, $p < .001$) significantly elevated credit scores (inclusive of right-to-sell scores), with a medium effect size (Cohen's $d = 0.62$, 95 % CI [0.4, 0.8]) in excess of our pre-registered smallest effect size of interest (Cohen's $d = 0.30$), and with a CLES of 68 % (i.e., revealing higher credit allocations in the hand-made condition 68 % of the time in repeated comparisons between randomly sampled observations, see Fig. 4). More credit was also attributed to Robin when making the paintings by hand ($M = 86.6$, 95 % CI [82.0, 91.3]) than when using either personalized AI ($\Delta M = 32.0$, 95 % CI [26.1, 37.9], $t(540) = 12.76$, $p < .001$; Cohen's $d = 1.34$, 95 % CI [1.1, 1.5], CLES = 85 %) or standard AI ($\Delta M = 48.6$, 95 % CI [42.0, 55.1], $t(539) = 17.46$, $p < .001$; Cohen's $d = 2.11$, 95 % CI [1.8, 2.4]; CLES = 91 %).

Overall, this demonstrates that the use of personalized AI models to create an artistic image can elevate credit attributions to the human user when compared to standard AI models. Still, a user of personalized AI receives less credit than a human artist who creates the painting entirely by hand. Thus, personalized AI use only partially bridges the “achievement gap” entailed by the use of generative AI in the visual arts domain.

3.4.2. Does AI authorship reduce aesthetic appreciation?

Consistent with H2 and the previous results of Study 1, we found that any AI use in making an artistic image (regardless of personalization) reduces aesthetic appreciation of the image compared to the same image described as being made “by hand” (i.e., solely by a human) (see Fig. 4). Robin's paintings received similar low aesthetic appreciation scores in the personalized condition ($M = 50.2$, 95 % CI [44.2, 56.1]) and standard condition ($M = 49.8$, 95 % CI [43.6, 56.1]), with no significant difference between the mean scores ($\Delta M = 0.3$, 95 % CI [-6.4, 7.1], $t(541) = 0.12$, $p = .99$). However, Robin's hand-made paintings ($M = 62.9$, 95 % CI [56.9, 68.9]) received significantly higher aesthetic appreciation scores than the paintings in either the personalized AI condition ($\Delta M = 12.7$, 95 % CI [6.5, 18.9], $t(541) = 4.82$, $p < .001$; Cohen's $d = 0.49$, 95 % CI [0.3, 0.7], CLES = 63 %) or the standard AI condition ($\Delta M = 13.1$, 95 % CI [6.2, 19.9], $t(541) = 2.88$, $p < .001$; Cohen's $d = 0.51$, 95 % CI [0.3, 0.7]; CLES = 63 %).

The similar magnitude of the decrease in aesthetic appreciation in both the personalized and standard AI conditions (compared to control) is striking. It shows that AI use of any kind in generating an artistic image dampens aesthetic appreciation of the image in human viewers, regardless of personalization. While this is consistent with findings on algorithm aversion in prior research on AI generated creative outputs (Bellaiche et al., 2023; Köbis & Mossink, 2021; Magni et al., 2023), we extend these findings to show that personalization does not mitigate such aversion. Furthermore, although the personalized AI described in the vignettes was “trained on an artist's own past paintings” (that were made by hand), this did not improve aesthetic appreciation scores in the same way as it improved credit attribution. Aesthetic assessments of artwork may thus involve a different set of evaluative judgments than those for credit assignment. We explore these distinctions further in our exploratory analyses.

3.4.3. Exploratory analysis

We pre-registered additional analyses, described below, of exploratory dependent variables added for Study 2: creativity, ability, effort, and art categorization (i.e., judgments that the image is truly art). We first report the means and standard deviations of these measures in Table 1, along with previously described results for credit and aesthetic appreciation. This is to show that, on visual inspection, the results for creativity, ability, and effort seem to follow the same basic pattern as

was previously found for credit attributions, whereas the art categorization judgments seem to follow the same basic pattern as was previously found for aesthetic appreciation. Based on this observation, we decided to present the pre-registered exploratory analyses in two separate sections: One for creativity, effort, and ability (mirroring the credit pattern), and one for art categorization (mirroring the aesthetic appreciation pattern), to facilitate interpretation. We also present the zero-order correlations between variables in Table 2.

3.4.4. Variables following the credit pattern

As pre-registered, we first regressed each of the four supplementary variables on condition as the predictor variable, with painting as a random intercept. The results for effort, ability and creativity are shown next (Fig. 5), with the results for art categorization shown in a subsequent section.

Effort. Robin was judged to have exerted more effort when using a personalized AI ($M = 47.6$, 95 % CI [45.2, 50.1]) than a standard AI ($M = 37.1$, 95 % CI [34.1, 40.0], $\Delta M = 10.6$, 95 % CI [6.0, 15.2], $t(541) = 5.43$, $p < .001$; Cohen's $d = 0.54$, 95 % CI [0.3, 0.8]; CLE = 65 %). However, when Robin made the painting by hand ($M = 72.0$, 95 % CI [69.5, 74.5]), effort attribution was higher than in either the personalized condition ($\Delta M = 24.4$, 95 % CI [20.2, 28.6], $t(541) = 13.63$, $p < .001$; Cohen's $d = 1.38$, 95 % CI [1.2, 1.6]; CLE = 83 %) or the standard condition ($\Delta M = 34.9$, 95 % CI [30.3, 39.6], $t(541) = 17.65$, $p < .001$; Cohen's $d = 2.13$, 95 % CI [1.9, 2.4]; CLES = 93 %).

Ability. Robin was judged as having more ability in the personalized condition ($M = 44.1$, 95 % CI [41.3, 47.0]) than in the standard condition ($M = 23.7$, 95 % CI [20.2, 27.2], $\Delta M = 20.4$, 95 % CI [15.0, 25.8], $t(541) = 2.29$, $p < .001$; Cohen's $d = 0.89$, CLES = 74 %). However, Robin's ability when making the painting by hand ($M = 78.6$, 95 % CI [75.7, 81.6]) was judged to be significantly higher than in either the personalized condition ($\Delta M = 34.5$, 95 % CI [29.6, 39.5], $t(541) = 16.38$, $p < .001$; Cohen's $d = 1.64$, 95 % CI [1.4, 1.9], CLES = 88 %) or the standard condition ($\Delta M = 54.9$, 95 % CI [49.5, 60.4], $t(541) = 23.55$, $p < .001$; Cohen's $d = 2.91$, 95 % CI [2.6, 3.2]; CLES = 96 %).

Creativity. Robin's creativity was also rated higher in the personalized condition ($M = 52.8$, 95 % CI [49.8, 55.9]) than in the standard condition ($M = 34$, 95 % CI [30.2, 37.7], $\Delta M = 18.9$, 95 % CI [13.1, 24.7], $t(541) = 7.65$, $p < .001$; Cohen's $d = 0.79$, 95 % CI [0.6, 1.0]; CLES = 72 %). However, Robin's creativity when the painting was made by hand ($M = 66$, 95 % CI [62.8, 69.2]) was judged as significantly higher than in either the personalized condition ($\Delta M = 13.2$, 95 % CI [7.8, 18.5], $t(541) = 5.82$, $p < .001$; Cohen's $d = 1.44$, 95 % CI [1.2, 1.7], CLE = 66 %) or the standard condition ($\Delta M = 32$, 95 % CI [26.1, 37.9], $t(541) = 12.78$, $p < .001$; Cohen's $d = 0.60$, 95 % CI [0.4, 0.8], CLES = 84 %).

What explains differences in credit? As noted previously, the effort, ability, and creativity findings seem to mirror those of the credit findings earlier described. Are attributions of credit thus linked in some way to perceptions of effort, ability, and/or creativity? To answer this question, we entered effort, ability, and creativity ratings as predictors in the model of credit attribution and used *subtractive model selection* to eliminate non-significant predictors of credit.

In the full model, effort did not independently predict credit attribution ($B = 0.10$, $t = 0.05$, $p = .06$). After removing effort, the ability-plus-creativity model (AIC = 3163.1) explained 71.9 % of the variance in credit attribution, providing substantially better fit to the data than the base model with the condition term only (AIC = 3763.7, $R^2 = .38$, $p < .001$). Ability ($B = 0.44$, $t = 9.52$, $p < .001$) and creativity ($B = 0.48$, $t = 11.31$, $p < .001$) significantly predicted credit attribution. When controlling for ability and creativity, the effect of manually creating art on credit attribution was positive, though substantially weaker ($B = 9.54$, $p < .001$ vs. $B = 48.6$ in the base model). However, the effect of AI

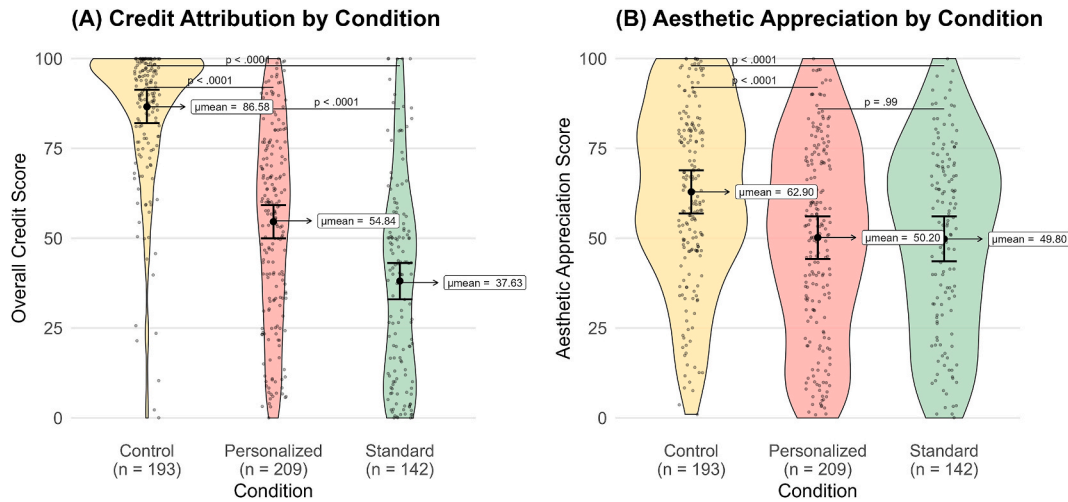


Fig. 4. Credit and Aesthetic Appreciation Scores. Violin plots showing credit (A) and aesthetic appreciation (B) scores by condition in study 2.

Table 1
Means and standard deviations for Study 2.

	Personalized		Standard		Control	
	M	SD	M	SD	M	SD
Credit	54.84	27.43	37.63	28.05	86.58	18.94
Effort	47.64	20.08	37.06	18.52	72.01	14.68
Ability	44.14	24.23	23.71	21.10	78.64	17.06
Creativity	52.84	23.41	33.97	24.30	66.0	20.52
Aesthetic appreciation	50.05	27.39	50.04	26.51	63.07	25.12
Art categorization	53.73	25.98	51.02	28.53	80.81	17.80

personalization on credit was rendered non-significant ($B = -0.80$, $t = -0.41$, $p = .68$). These results indicate that perceptions of the artist's ability and creativity fully explained the effect of personalization on credit – relative to standard AI use. Furthermore, controlling for perceptions of ability and creativity left only a small preference for manually created art over AI-generated art.

3.4.5. Variable following the aesthetic appreciation pattern

Art Categorization. Unlike the other secondary variables, ratings of whether the painting could be categorized as true art dropped equally in both AI conditions (Fig. 6). The perceived status of the painting as true art decreased by a similar amount when Robin used either a personalized AI ($M = 53.73$, 95 % CI [50.4, 57.0]) or a standard AI ($M = 51.0$, 95 % CI [47.0, 55.0]), with a non-significant difference between these two ($\Delta M = 2.7$, 95 % CI [-3.5, 8.9], $t(541) = -11.15$, $p = .56$). Paintings were more strongly considered to be art when Robin made them by hand ($M = 80.8$, 95 % CI [77.4, 84.2]) as compared to when they were made using personalized AI ($\Delta M = 27.1$, 95 % CI [21.4, 32.8], $t(541) = 11.22$, $p < .001$; Cohen's $d = 1.21$, 95 % CI [0.9, 1.4]; $CLES = 80$ %) or standard AI ($\Delta M = 29.8$, 95 % CI [23.5, 36.1], $t(541) = 11.15$, $p < .001$; Cohen's $d = 1.30$, 95 % CI [1.1, 1.5]; $CLES = 80$ %).

Again, we sought to understand whether art categorization helped to

Table 2
Correlation table for study 2.

	(1)	(2)	(3)	(4)	(5)	(6)
(1) Credit	–					
(2) Effort	.70***	–				
(3) Ability	.81***	.80***	–			
(4) Creativity	.76***	.65***	.74***	–		
(5) Aesthetic appreciation	.35***	.29***	.39***	.42***	–	
(6) Art categorization	.70***	.58***	.66***	.64***	.49***	–

Note. * $p < .05$, ** $p < .01$, *** $p < .001$.

explain differences in aesthetic appreciation across conditions. To answer this question, we entered art categorization as a predictor in the model of aesthetic appreciation.

This model⁵ (intercept: $B = 24.2$, $t(540) = 6.36$, $p < .001$) revealed that art categorization predicted aesthetic appreciation ($B = 0.48$, 95 % CI [0.40, 0.56], $t(540) = 11.40$, $p < .001$), when controlling for condition. Paintings generated using personalized AI ($\Delta M = -0.0$, 95 % CI [-6.2, 6.2], $t(540) = 0.00$, $p = 1$) and standard AI ($\Delta M = -1.3$, 95 % CI [-8.1, 5.5], $t(540) = 0.44$, $p = .66$) no longer received lower aesthetic appreciation ratings than did hand-made paintings. This model explained 23 % of the variance, providing substantially better fit to the data than the base model with the condition term only ($R^2 = .049$). This analysis suggests that the aesthetic preference for hand-made art over AI-generated art is accounted for by a greater tendency to view the former as qualifying as art.

3.5. Discussion of secondary analyses

Our additional measures suggest that while personalization improves perceptions of the artist's contribution (effort, ability, creativity) alongside credit attribution, it fails to enhance either aesthetic appreciation or art categorization (i.e., the perception that the image really is art).

What explains this? Previous research has shown that 'art' is a dual character concept (Liao et al., 2020), meaning that it possesses both descriptive and evaluative criteria for its application (for research on dual character concepts, see Knobe et al., 2013; Leslie, 2015; Liao et al., 2020). This suggests that AI-generated images, regardless of their technical achievement, may fail to meet the full evaluative criteria for 'art.' The responses might thus reflect a judgement that true art must embody a set of values that is out of the reach of AI to instantiate. This would be compatible with previous research, which suggests that perceived aesthetic value (Kamber et al., 2018; Mikalonytė & Kneer, 2024) and judgments of personal preference (i.e., liking; Pelowski et al., 2017) are among the most important predictors of an object being categorized as art, over and above its mere technical constitution.

⁵ As noted previously, a look at the means across conditions (Table 1) shows that effort, ability, and creativity follow the same patterns as credit (they increase between standard and personalized conditions). However, there is no corresponding increase in aesthetic appreciation as would be expected if the three former variables mediated the effects of condition on aesthetic appreciation. Thus, we decided not to include effort, ability and creativity in this model.

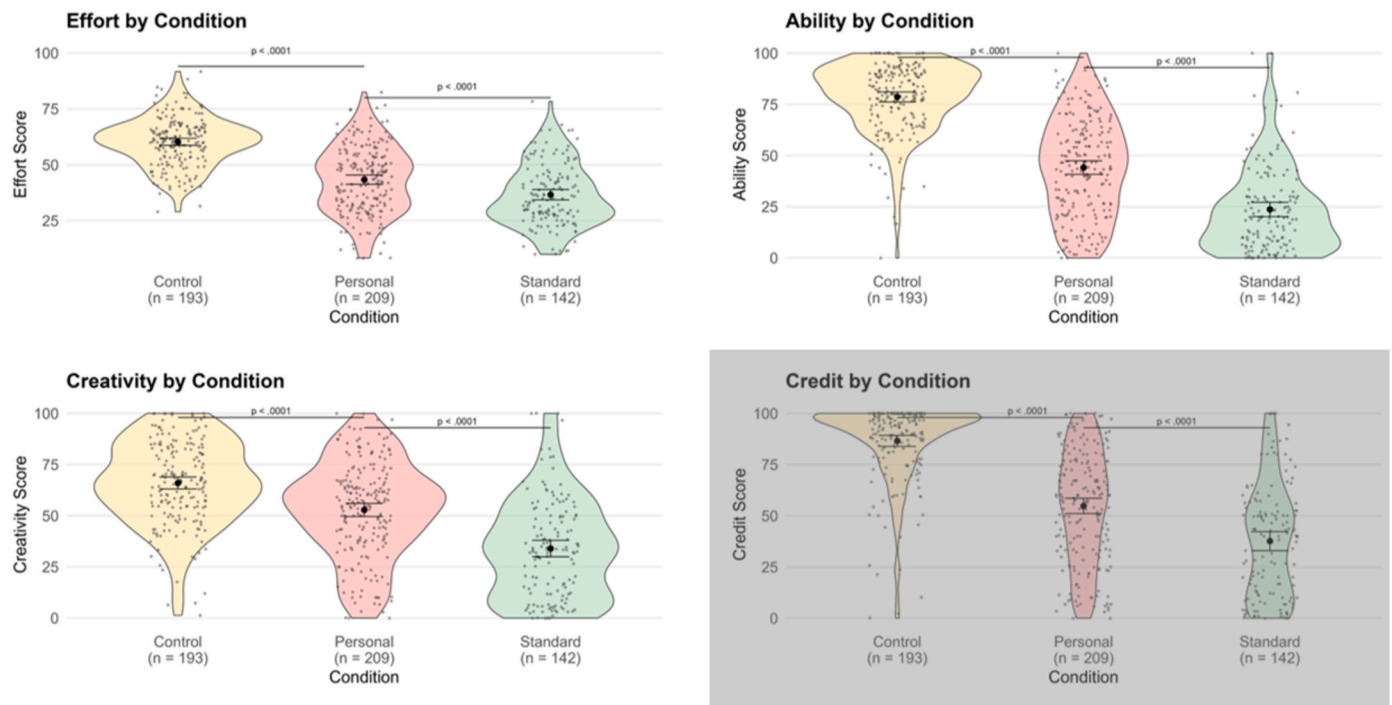


Fig. 5. Effort, Ability and Creativity. Exploratory variables following patterns similar to credit. *Note:* The previously reported credit results are reproduced here and are highlighted in grey for reference.

Taken together with our pre-registered analyses, these findings suggest that credit attribution for artworks involves the evaluation of a different set of features than the ones that contribute to aesthetic appreciation – features that personalization can improve, such as effort, ability, and creativity. These same criteria do not apply to aesthetic appreciation or categorization of AI-generated works as true art. Instead, our findings imply that there is some other set of evaluative criteria for classifying an image as art that AI cannot meet despite being trained on the works of a specific human artist.

4. Discussion

Our findings reveal a fundamental disconnect in how people evaluate AI-assisted art: While personalizing AI models can help artists maintain (what others perceive as) legitimate claims of authorship and associated commercial rights, it cannot overcome the inherent barriers to aesthetic appreciation of AI-generated art, or even the relatively lowered perceptions that AI-generated images are, in fact, art. This finding challenges assumptions about the relationship between perceived artistic effort and aesthetic value.

In Study 1, we found that personalizing an AI model improved both credit attribution and the artist's perceived right to sell compared to standard AI use. Study 2 replicated and extended these findings, showing that personalization enhanced not only credit attribution (including perceived commercial rights) but also perceptions of effort, ability, and creativity. This suggests that personalization can help at least partially bridge the “achievement gap” typically associated with AI assistance, consistent with earlier findings in relation to written work (Earp, Porsdam Mann, et al., 2024), extended here for the first time to the visual arts. In this domain, we also found through exploratory analyses that the increase in credit that was caused by personalization was likely driven by an increased perception of creativity and ability on the part of the artist, rather than effort *per se*. This exploratory finding should be followed up in future work.

Across both studies, however, paintings described as having been generated using personalized AI received similar aesthetic appreciation

ratings, and perceptions of being truly art, to those created with standard AI. Both variables, across both types of AI, were rated significantly lower than identical images described as hand made.

This disconnect between contribution and appreciation challenges both traditional philosophical theories of aesthetic evaluation and previous empirical research. Contextualism in philosophical aesthetics (Bradford, 2015) correctly predicts that knowledge about the creator's identity and the creative process can influence judgments of aesthetic value. However, the extent of the creator's achievement embodied in an artwork, as represented by the creative process reflecting their previous artistic effort, did not increase aesthetic appreciation. While research shows that people typically use cues like brush strokes as heuristics for effort in aesthetic judgments (Chamberlain et al., 2018; Leder et al., 2012), our findings suggest that changes in contextual information about the painting — whether it was described as having been created using a general or personalized model — did not translate into enhanced aesthetic appreciation when AI is involved. This suggests that only the creator's identity and methods of creation, but not the level of their personal artistic achievement, is considered relevant to aesthetic value.

One explanation comes from our findings' alignment with previous work on algorithm aversion in art appreciation (e.g., Chamberlain et al., 2018), showing that knowledge of AI involvement reduces perceived aesthetic value. Crucially, participants in the control condition saw the very same paintings as in the AI conditions, all of which were generated by an AI system, Dall-E, in November 2023. Therefore, the lower aesthetic appreciation and relative denial of AI art as “true art” that we found in both AI conditions did not appear to be caused by any perceptual property of the work itself. The reduction of aesthetic value for AI-generated artworks has also been reported in previous work on algorithmic aversion towards AI art (Hong & Curran, 2019), suggesting there may be unique evaluative criteria that are applicable to AI.

In our previous discussion (Study 2), we consider art as a dual character concept that has both descriptive and evaluative criteria (Knobe et al., 2013; Liao et al., 2020). We suggest that an intriguing explanation for our results may be that, regardless of AI's technical achievement, AI-generated images fail to meet the full evaluative

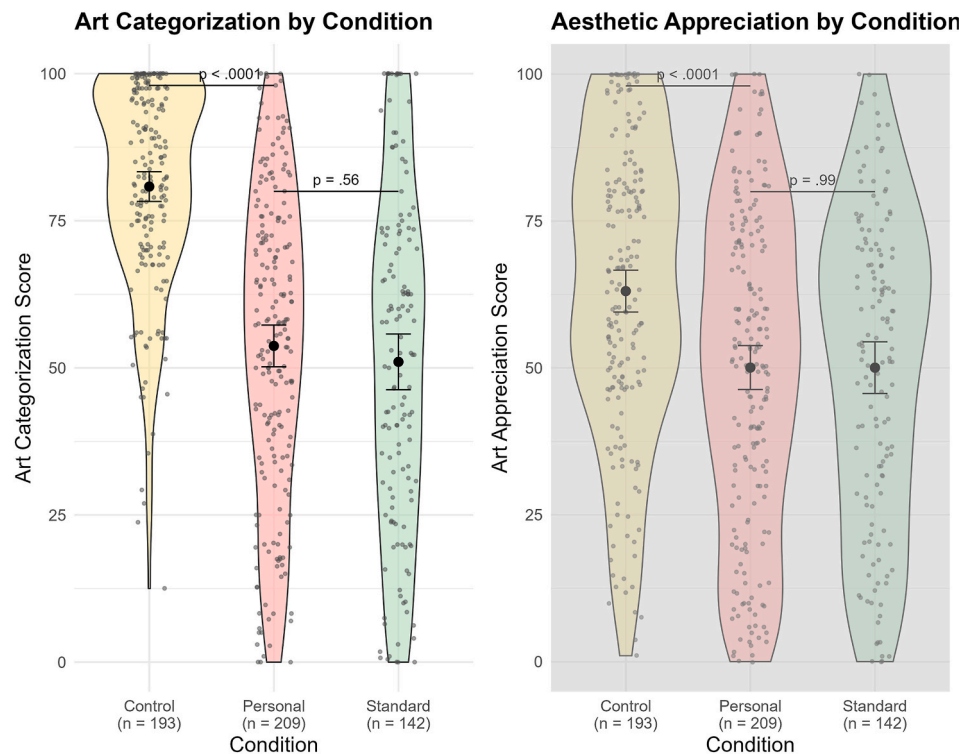


Fig. 6. Art Categorization and Aesthetic Judgment. AI personalization fails to enhance judgments of art categorization or aesthetic appreciation. *Note.* the previously reported aesthetic appreciation results are reproduced here and are highlighted in grey for reference.

criteria for 'art'. Moreover, algorithm aversion may play a role in this (negative) evaluation, offering a possible explanation of why AI-generated art, despite its being a reflection of computational or technical prowess, may not be considered aesthetically valuable or "true art" by human evaluators. Thus, the lower aesthetic evaluations of AI art may be a product of how people tend to think about AI outputs more generally.

These findings have important implications for the future of AI in art creation. While personalization offers a promising path for maintaining artistic credit and authorship claims, overcoming resistance to AI-generated art's aesthetic value may require addressing deeper questions about the nature of creative agency.

This work has some limitations. Firstly, most participants did not have any art expertise or extensive art experience – an art naive population such as this may have provided different responses to art experts, whose aesthetic value ratings of art have been documented to be higher than ratings from non-expert populations (Leder et al., 2012; Lindell & Mueller, 2011; van Paasschen et al., 2015). Secondly, while we pre-tested the paintings for quality, these were digital (rather than physical paintings). It is possible that participants would have provided different ratings on aesthetic value and art categorization judgments if these works had been physical (Chamberlain et al., 2018; Magni et al., 2023; Mikalonytė & Kneer, 2022; Zhou & Kawabata, 2023). Finally, our sample was limited to UK participants. Evidence that algorithm aversion may be weaker in non-Western samples (Chu & Liu, 2023; Liu et al., 2025) points toward the possibility of cultural differences in the aesthetic appreciation and categorization of AI-assisted art. The current work therefore invites cross-cultural replication.

5. Conclusion

This study extends our understanding of how personalization affects perceptions of AI-generated content from written text (as in Earp, Porsdam Mann, et al., 2024) to visual art, providing the first empirical investigation of personalized AI in the context of visual art creation.

These results may have implications for artists, policymakers, and technology developers, as well as for scholarship on algorithmic aversion and the psychology of art. As AI becomes increasingly integrated into creative processes, understanding how personalization affects perceptions of authorship and artistic value will be crucial for developing ethical guidelines, copyright policies, and AI tools that support rather than undermine artistic practice. While our findings suggest that personalization may offer a partial solution to the "achievement gap" in credit associated with AI use in creative domains, they also indicate that reconciling AI assistance with traditional notions of aesthetic value remains a challenge. This work thus provides a foundation for future research on human-AI collaboration in art and may inform ongoing discussions about intellectual property rights and art policy in the age of AI.

CRedit authorship contribution statement

Maryam Ali Khan: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Elzé Sigutė Mikalonytė:** Writing – review & editing, Supervision, Methodology, Data curation, Conceptualization. **Sebastian Porsdam Mann:** Writing – review & editing, Methodology, Data curation, Conceptualization. **Peng Liu:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization. **Yueying Chu:** Methodology, Investigation, Data curation, Conceptualization. **Mario Attie-Picker:** Writing – review & editing, Methodology, Conceptualization. **Mey Bahar Buyukbabani:** Writing – review & editing, Methodology, Investigation, Data curation, Conceptualization. **Julian Savulescu:** Writing – review & editing, Supervision, Conceptualization. **Ivar R. Hannikainen:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Brian D. Earp:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization.

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Declaration of competing interest

Sebastian Porsdam Mann serves as a member of the advisory board for AminoChain Inc. Julian Savulescu is a Bioethics Committee consultant for Bayer and serves as Bioethics Advisor to the Hevolution Foundation. The remaining authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.techsoc.2025.103055>.

Data availability

Our data and code is available on the OSF which is linked.

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